A Atlas One

NAIC 2026 · ITEM 2

N-ATLaS Integration Evidence

How Atlas One integrates directly with the N-ATLaS model and API, and the live proof that every result comes from genuine N-ATLaS.

Product	Date	Live	Source
Atlas One v1.0	12 October 2026	atlas-one...workers.dev	github.com/agapemiteu/atlasone

Direct
serves NCAIR1/N-ATLaS itself; no other model or provider

e2944769
pinned model revision on every response

24 s
cold start, measured (container stopped → online)

27.6 tok/s
Hausa generation on the GPU

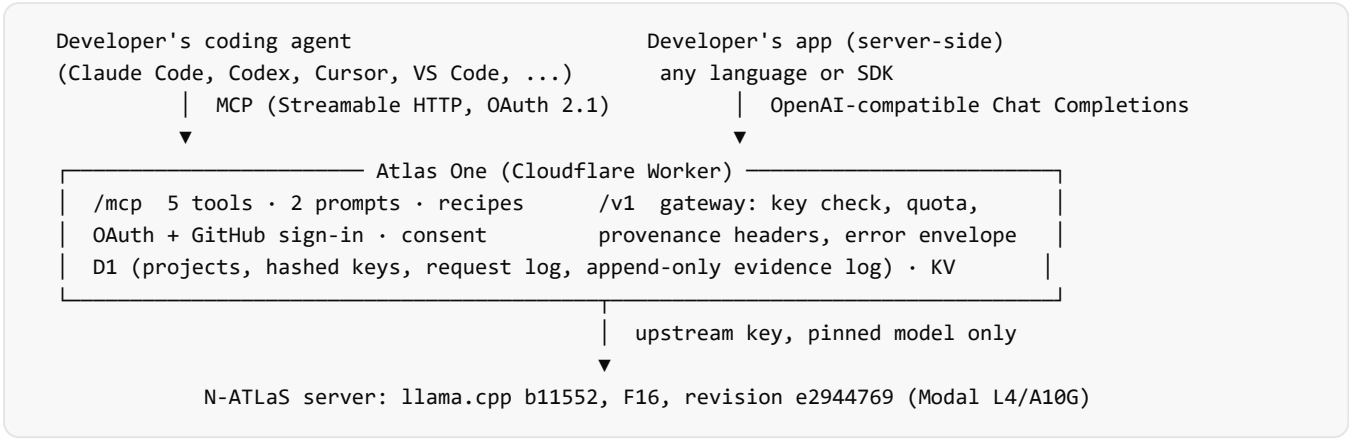
1. What Atlas One is

Atlas One is developer infrastructure for N-ATLaS (NAIC Problem Statement 1). It has two interfaces over one model:

- **An MCP server** (`https://atlas-one.collins-coordinator-worker.workers.dev/mcp`) that lets any coding agent discover N-ATLaS, create a project and key, integrate it into an app, run genuine inference and **prove** that the app's own route reached the model.
- **An OpenAI-compatible API** (`https://atlas-one.collins-coordinator-worker.workers.dev/v1`) that apps call in production, from any language or SDK.

2. The model integration

ITEM	VALUE
Model	`NCAIR1/N-ATLaS` (Llama-3 8B fine-tune; English, Hausa, Igbo, Yoruba)
Pinned revision	`e294476928aca9030e924ca27bb8e085e8581273`
Precision	F16 GGUF converted from the original BF16 weights with llama.cpp's converter; no quantisation
Runtime	llama.cpp `b11552` (CUDA), OpenAI-compatible server, 4 slots × 4,096 tokens, the model's own chat template
GPU	Modal: one NVIDIA L4 or A10G (24 GB), falls back between the two; scales to zero. Fallback provider: Kaggle 2× T4
Settings	From the N-ATLaS model card, applied server-side: repetition penalty 1.12, temperature 0.1, current date



The gateway only ever forwards to a registered N-ATLaS server. A server can register only if it is healthy, accepts the gateway's private upstream key and serves `n-atlas-8b`; registration is limited to the active provider's hosts. `package.json` contains no other model provider, and the only model call in the code (`src/lib/inference.ts`) goes to that server.

3. Provenance on every response

HEADER	MEANING
`x-atlas-request-id`	Unique id, also written to the request log and, for verified apps, to the signed evidence log
`x-atlas-weights`	Exact model and revision, `NCAIR1/N-ATLaS@e294476928aca9030e924ca27bb8e085e8581273`
`x-atlas-runtime`	`llama.cpp@b11552 f16`
`x-atlas-upstream-ms`	Time spent in the model server

App verification. `atlas_test` issues a single-use nonce; the app's own route forwards it as `x-atlas-verify`; `verify_app` returns `verified: true` only if a request carrying that nonce reached N-ATLaS through the developer's project. A signed receipt (JWT, EdDSA) records it and anyone can check it against the public key at `https://atlas-one.collins-coordinator-worker.workers.dev/.well-known/jwks.json`.

4. Live results

CHECK	REAL N-ATLAS OUTPUT	REQUEST ID	TIME
Gateway, Hausa chat	Babban birnin Najeriya Abuja ne.	`req_f48ae97d5398433ba507939d`	969 ms
MCP atlas_test, Yoruba	Olú ìlú Nàìjíríà ni Abuja.	`req_7b2f7b28977c40f18a8551d2`	834 ms
MCP verify_app (app route reached N-ATLaS)	verified: true	`req_5ac15decbba847578dd3150d`	
Structured output (JSON Schema), Hausa review	parseable JSON; sentiment negative; faithful English summary	`req_e77d8a9a9b4647799d092fb5`	
Structured output, medicine label → dosing JSON + Hausa	times_per_day 3, days 7, after_food true; Hausa explanation	`req_a6cfb6957a9c428d8605d95b`	9.1 s
Function calling (tools)	model called track_order({"order_id": "KF-1042"})	`req_b4714f8393c04836aa6dbaa3`	2.7 s
Function calling, answer from tool result	answered in Hausa from the result	`req_c20473f1dfdc4683b860865c`	2.0 s

CHECK	REAL N-ATLAS OUTPUT	REQUEST ID	TIME
Throughput on the GPU	191 Hausa tokens at 27.6 tokens/s	`req_5eba2c98f8814bc0acddc710`	6.9 s
Demo build: empty folder → app (video)	verified: true	`req_c373e5bee04545d4bf9b28c7`	6 min 31 s end to end

Language coverage (an external tester's own probes)

LANGUAGE	OUTPUT	REQUEST ID	LATENCY
Hausa	Babban birnin Najeriya Abuja ne.	`req_b8915c173c7a407087f0bb23`	1,049 ms
English	The capital of Nigeria is Abuja.	`req_6316e597b6854ccda346b2f6`	822 ms
Igbo	Isi obodo Naijiria bụ Abuja.	`req_b4c8fa1d784848e19657a890`	1,025 ms
Yoruba	Olú ìlú Nàìjíríà ni Abuja.	`req_39e3bfd3dc9f475db7b3337f`	1,264 ms

5. Full model capabilities through the API

CAPABILITY	HOW	CHECKED LIVE
Chat, multi-turn, four languages	messages array; reply language in the system message	yes
Streaming	`stream: true` (Server-Sent Events)	yes
Structured output	`response_format` json_object / json_schema, grammar-enforced	yes
Function calling	`tools`, `tool_choice`, `tool_calls`, role tool	yes
Sampling controls	temperature, top_p, seed, stop, penalties, max_tokens ≤ 1024	yes

6. Honest failure

Atlas One never substitutes another model. If N-ATLaS is asleep, starting or unreachable, every interface returns a structured error (`BACKEND_WARMING` with `retry_after_s`, `MODEL_OFFLINE`, `BACKEND_BUSY`, `INFERENCE_TIMEOUT`, `INVALID_OUTPUT`) with a request id. Output quality is reported as observed: an external tester found grounding and language-adherence issues in a customer-support use case (see the Real-World Validation report); Atlas One reports them rather than hiding them.

7. Check it yourself

1. `curl https://atlas-one.collins-coordinator-worker.workers.dev/v1/models -H "authorization: Bearer <key>"` lists `n-atlas-8b` with its revision. Get a key at `https://atlas-one.collins-coordinator-worker.workers.dev/access` (sign in with GitHub, Create API key).
2. Send any chat completion and read the `x-atlas-weights` header.
3. `node scripts/verify-evidence.mjs` in the repository verifies the signed tester log and receipts against the live server.